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Sub : Chemistry Assignment 13

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Difference between E1 and E2

S.N

E1

E2

Unimolecular elimination

Bimolecular elimination

Two step reaction

one step reaction

forms a carbocation intermediate

No carbocation formed

Rearrangement can occur

No rearrangement

Favored by weak bases + heat

Favored by strong bases

Rate depends on one molecule

Rate depends on substrate + base

Can give Zaitsev product

usually gives Zaitsev,

but bulky favor Hofmann product.

example - tertiary alkyl halides - ex -

primary / secondary / tertiary with strong base.

Q.2 Zaitsev vs Hofmann product.

- In elimination (E2) reaction, a base removes a β hydrogen while the leaving group leaves, forming a double bond.

- Zaitsev product - double bond forms at the more substituted position giving the more substituted and more stable alkene with small bases like NaOH , or EtONa it is generally the major product.

Hofmann product: Double bond forms at the less substituted posits it gives a less substituted positions. It gives a less substituted alkene with a bulky base like t -BuOK the less hindered β -hydrogen is removed, so the Hofmann product becomes major.

example from the image

$\text{NaOH} / \text{EtONa} \rightarrow \text{Zaitsev major}$

$t\text{-BuOK} \rightarrow \text{Hofmann major}$

classification